MENG2111: Reading Comprehension

Read the passage below and answer the following questions.

The Electric Car Revolution: Is it Realistic or Optimistic?

Electric vehicles have arrived. With technology led by Tesla, and all of the world's major car manufacturers following along behind, electric vehicles are now a common sight on the roads of most developed countries. Yet the situation in less developed countries is rather different; the only African country to have started the change to electric vehicles is South Africa and even there, electric vehicles still account for less than 0.01% of the total number of cars on the roads. In South America, the situation is better, with all Latin American countries beginning the move towards electric vehicles, particularly Colombia which, in 2020, had a third of the continent's total electric car fleet. In Russia, the wealthy are investing in imported electric cars, but no electric cars are yet manufactured locally, while in India the government is promoting the purchase of electric vehicles with tax exemptions and other incentives. So electric cars have arrived, and their share of the market is increasing almost worldwide.

Does this mean, therefore, that the world is on track to phase out the use of petrol-driven vehicles in less than thirty years? And does it mean that electric vehicles are the sustainable solution to our transport needs for the second half of the century? Unfortunately, to the disappointment of some people, the answer to both of these questions has to be "no".

The massive development of electric vehicles can only be possible if two conditions are met. Firstly the expansion of electric vehicle manufacturing is dependent on the fragile ability of manufacturers to source vastly increased quantities of vital components and elements without which electric vehicles cannot operate; these include lithium, cobalt and "rare earths" such as neodymium and tantalum, as well as silicon chips which have already been in short supply since 2020. Secondly, few countries currently have electricity grids that are anywhere near being able to cope with the huge increase in demand for electricity that will accompany any rapid growth in electric vehicle ownership. Without adequate supplies of all the vital ingredients of electric motors and batteries, or without power supplies that are able to provide the electricity needed to recharge millions of electric batteries every day (as well as supplying the current we need for everything else, such as lighting, heating, trains and electric devices), the electric car revolution will run up against insoluble problems.

Governments and vehicle manufacturers are fully aware of these issues, but the consensus among policy-makers seems to be that somehow technology will come up with the answers, as it often has in the past. Analysts also predict that changing social attitudes and environmental awareness will lead to a reduction in private vehicle use and a fall in the numbers of vehicles on the roads. This prediction is likely to be right, though not necessarily for those reasons alone; any shortage of essential components will force up the cost of electric vehicles, and any shortage of battery recharging facilities or capacity will discourage people from buying electric vehicles, leading to a fall in the number of vehicles on the roads.

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The Electric Car Revolution: Is it Realistic or Optimistic? (Continued)

Ultimately the success of the transition to electric powered vehicles will depend on advances in technology in three fields; the weight of batteries, the amount of power that they can produce, and the speed at which they can be recharged...or exchanged.

It may surprise you to learn that electric vehicles are not a new idea; indeed, at the start of the automobile age in the late 19th century, America had as many electric cars as gas-driven cars, and New York's biggest taxi company used electric vehicles. In order to keep their taxis on the road when the batteries ran low, they set up battery points at strategic locations where instead of recharging their batteries, drivers would just exchange them for fully recharged ones, a process that took no longer than refilling with gasoline.

Thus battery exchange stations, rather than battery recharging points, may perhaps solve the problem of slow recharging times which currently prevents owners taking their electric vehicles on long trips. Yet battery exchange is not an option with today's large heavy batteries. While the latest generation of Lithium-ion batteries are almost twice as efficient as the batteries being used just five years ago, they remain big and heavy. Before batteries can become easy to exchange, a quantum leap in battery technology is needed, one which will allow batteries to store much larger amounts of electricity in much smaller and lighter units. Teams of top electrical engineers in universities and private laboratories worldwide are working on battery technology, and progress has been rapid. New types of lithium batteries are being developed, but the radical discovery that will revolutionize battery design is still to be made.

As for the availability of clean renewable electricity in sufficient quantities to cope with demand from all the world's electric vehicles, progress in this direction is already underway. Ideally, notably in hot countries and outside cities, recharging points will be autonomous, generating their own electricity from solar panels and wind or water turbines and storing it either mechanically or in high-powered batteries. The technology already exists, and a California company, Beam Global, recently installed 30 solar-powered recharging stations in sunny parts of the state.

So are people who believe in the ability of technology to solve all our problems being realistic, or over-optimistic? And will other problems such as environmental issues and the availability of vital materials throw a spanner in the works? Will all the world's countries be able to complete the electric vehicle revolution, or will the world's poorer nations get left behind....yet again? For the time being, progress is upwards, but can it continue? In terms of volume, the electric vehicle revolution has only just got underway.

(967 words)

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Based on the information in the passage, answer:

TRUE (T) - if the statement agrees with the information

FALSE (F) - if the statement disagrees with the information

NOT GIVEN (N) - if there is no information given in the passage

1. Electric cars are only possible today because of technology.

Ans: _____

2. Progress is underway by scientists to develop many potential solutions.

Ans: _____

3. Petroleum-based automobiles will be phased out in 30 years.

Ans: _____

4. There are still many potential development difficulties.

Ans: _____

Based on the passage, answer the following questions in complete sentences. Give detailed explanations to support you answers.

5. What are the two production development difficulties mentioned for electric cars?

6. What is a possible solution to overcome the shortcomings of the battery for electric cars?